Who actually decides?

Measuring voting-power concentration in Compound governance

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Token voting looks decentralised

- DAOs govern protocols holding billions of dollars by token-weighted on-chain votes.
- Compound: anyone holding COMP can vote, directly or by delegating.
- The usual decentralisation metrics are **token share** and the Gini coefficient.

But token share is not decision power.

- A 49% holder facing one 51% holder has *no* power; facing 51 holders of 1% each, it has almost all.
- Power depends on *who else* holds weight, the *quota*, and *delegation*.

Question: how far apart are nominal share and effective power in a real DAO, and how often did a handful of addresses decide outcomes?

What we did

1. Rebuilt **every** Compound GovernorBravo proposal (351) and vote (14,396) from raw Ethereum logs.
2. Reconstructed each delegate's voting weight at each proposal's snapshot by **replaying all 2.33M COMP transfers and 16k delegation events**.
3. Implemented delegation resolution, **Banzhaf** and **Shapley–Shubik** indices by dynamic programming ourselves (no index library).
4. Compared token share against both indices, and the indices against what actually happened in each vote.

Everything is reproducible from one config file and five commands; every number on these slides is generated by a script.

Weighted voting games

- Players $N = \{1, \ldots, n\}$ with weights $w_i \geq 0$ and a quota q.
- A coalition S **wins** iff $w(S) = \sum_{i \in S} w_i \geq q$. Written $[q;\, w_1, \ldots, w_n]$.
- Player i **swings** for $S \subseteq N \setminus \{i\}$ if S loses but $S \cup \{i\}$ wins:

$$q - w_i \;\leq\; w(S) \;<\; q.$$

Example $[51;\, 50, 30, 20]$: winning coalitions $\{1,2\}, \{1,3\}, \{1,2,3\}$.
Player 1 holds 50% of the weight but is in every winning coalition.

Two power indices

Banzhaf: how often is i a swing, over all coalitions of the others?

$$\psi_i = \frac{\#\{S \subseteq N \setminus \{i\} : q - w_i \le w(S) < q\}}{2^{n-1}}, \qquad \beta_i = \frac{\psi_i}{\sum_j \psi_j}.$$

Shapley–Shubik: in a random ordering of players, how often is i the one whose arrival first reaches q?

$$\varphi_i = \sum_{\substack{S \subseteq N \setminus \{i\} \\ q - w_i \le w(S) < q}} \frac{|S|!\,(n-1-|S|)!}{n!}.$$

$[51; 50, 30, 20]$	player 1	player 2	player 3
token share	0.50	0.30	0.20
Banzhaf β	0.60	0.20	0.20
Shapley–Shubik φ	0.667	0.167	0.167

(The project brief listed $\beta = (0.50, 0.25, 0.25)$; enumeration gives 3, 1, 1 swings, so $(0.6, 0.2, 0.2)$.)

Delegation in COMP

- A holder's tokens vote only if it **delegates**, to itself or to another address.
- Delegation is **one-hop**: if $A \to B$ and $B \to C$, then A's tokens count for B, and B's own tokens for C.
- Undelegating ($\to$ `0x0`) removes the votes; undelegated tokens count for nobody.
- Voting weight at a proposal $= \sum$ balances delegated to you, at its `startBlock`.

Our resolver implements this rule, plus a *transitive* variant for sensitivity:

- follow $A \to B \to C \to \cdots$ to a self-delegated or non-delegating address;
- **cycle rule:** addresses on a cycle are treated as self-delegated; a chain entering a cycle stops at the first cycle address.

Pipeline

Stage	What it does
`data.fetch`	`eth_getLogs` for 8 event types over blocks 9.6M–22.0M, adaptive range splitting, cached to `data/raw/` (175 MB)
`data.process`	ABI-decode; replay transfers + delegations in one pass over all snapshots; write CSVs
`power/`	delegation resolver, Banzhaf DP, Shapley–Shubik DP, flip sets, gap metrics
`analysis.run_all`	all tables, figures, and the numbers on these slides

- Scope: Compound GovernorBravo, proposals 43–393 (281 executed, 33 defeated, 37 canceled).
- Players: every address with positive voting weight at the snapshot (median 4,912 per proposal).
- Quota: pass iff for $>$ against and for $\geq$ 400k COMP $\Rightarrow$ $q = \max(\lfloor W/2 \rfloor + 1,\ Q)$.

The data is exact

We never trust a third-party API for weights; instead we check our reconstruction against the chain itself.

Check	Mismatches
Our delegate weights vs on-chain checkpoints (all delegates, 351 snapshots)	**0**
Our weight vs the weight of each of the 14,396 `VoteCast` events	**0**
Our for/against/abstain tallies vs `proposals(id)`	**0**
Our outcome label vs `state(id)`	**0**

This validates the delegation resolver and the snapshot rule (events in block $\leq$ `startBlock`) end to end.

Computing the indices: dynamic programming

Banzhaf. $p[x] = \Pr(w(S) = x)$ for S a random subset; add a player of weight v:

$$p'[x] = \tfrac{1}{2}\big(p[x] + p[x - v]\big), \qquad x < q.$$

Shapley–Shubik. $P[k, x] = \Pr(w(S) = x)$ for S a random k-subset of the m players so far:

$$P'[k, x] = \frac{(m + 1 - k)\, P[k, x] + k\, P[k - 1, x - v]}{m + 1},$$

$$\varphi_i = \frac{1}{n} \sum_{k=0}^{n-1} \Pr_k\big(q - w_i \le w(S) < q\big).$$

- Convex combinations: no overflow, even for thousands of players (raw counts reach $\binom{n}{n/2}$).
- Leave-one-out by **divide and conquer** over distinct weights: only additions, no unstable "removal".
- Weights rounded to $W/20000$; dummies carry $\le 0.1\%$ of weight. Changing the resolution moves any index by ≤ 0.00016.

Tests (280, all passing)

Indices

- known example $[51; 50, 30, 20]$ to 10^{-9}
- dummy players get 0
- equal weights $\Rightarrow$ equal indices
- Shapley–Shubik sums to 1
- DP = brute force on 40 random games with $n \leq 12$
- DP = permutation definition ($n \leq 7$)

`pytest` runs in about 2 seconds.

Delegation & outcomes

- chains, cycles, self-delegation, undelegation, re-delegation
- same-block ordering by log index
- snapshot unaffected by later events
- one-pass replay = direct recomputation
- minimal flip set = brute force on 200 random votes

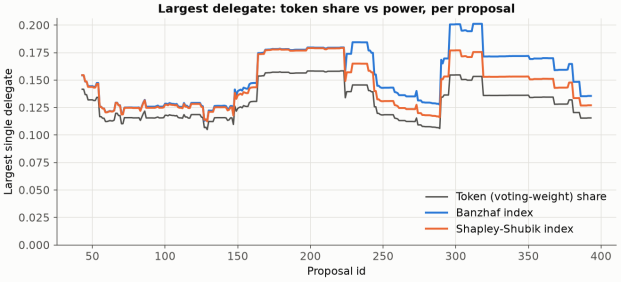
A few delegates hold most of the weight

Median over 351 proposals:

	largest delegate
token share	13.1%
Banzhaf	14.9%
Shapley–Shubik	14.3%

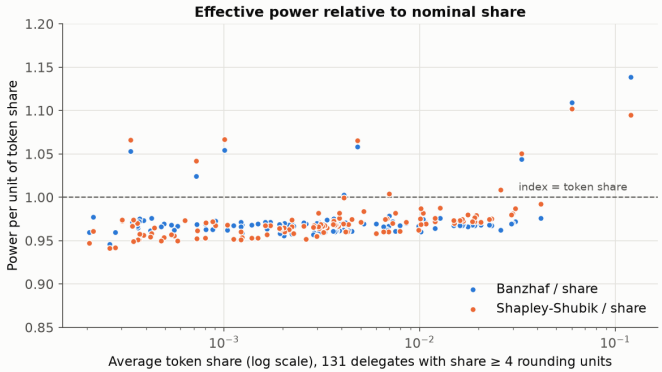
Top 5: 38.2% of weight,
39.6% / 40.6% of power.

Of 4,912 players, only ∼76 are not dummies.

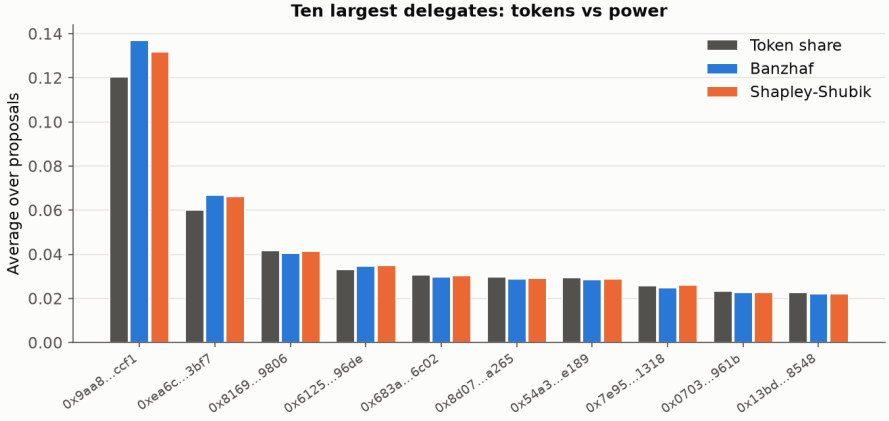


Nominal vs effective: the largest are amplified

- The 1–2 largest delegates get $\approx 1.1\times$ power per unit of tokens.
- Nearly everyone else is compressed to $\approx 0.96\times$.
- Largest single gap: $+4.8$ points (Banzhaf), $+2.2$ points (Shapley–Shubik), both for the top delegate.



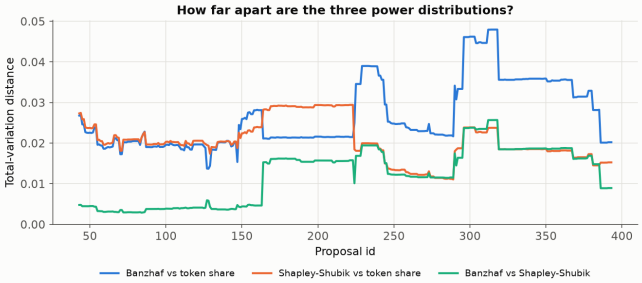
Ten largest delegates



Averaged over all proposals. Addresses are shown, not identities: one entity may control several.

Banzhaf vs Shapley–Shubik: same order, different sizes

- Spearman rank correlation is 1.000 in *every* proposal.
- That is a theorem: in a weighted game, a heavier player is always at least as powerful under both indices.
- The real difference is in magnitude: total-variation distance median 0.015 (max 0.026).
- Banzhaf favours the top delegate more after proposal ∼225.

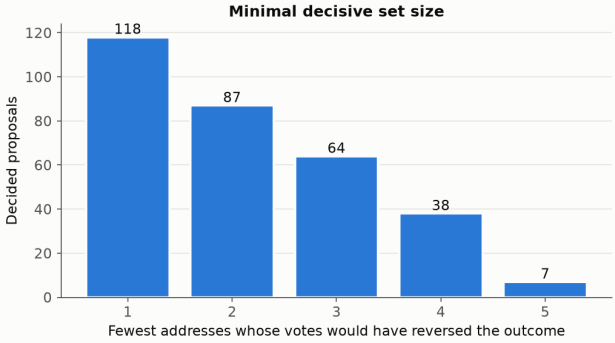


From power to outcomes: how many addresses decided?

Flip set: players who, changing only their own votes, reverse the result (with quorum).

Of 314 decided proposals:

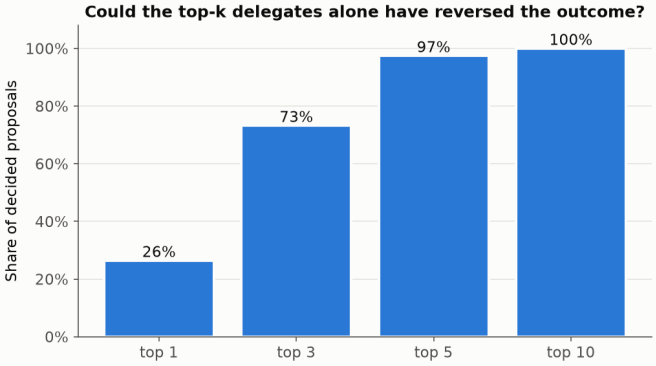
- **118** could be reversed by **one** address
- median minimal set: 2; never more than 5
- 28 of 33 defeated proposals: one large non-voter could have passed them



Could the top-k have reversed it alone?

k	share of outcomes
1	26.4%
3	73.2%
5	97.5%
10	100.0%

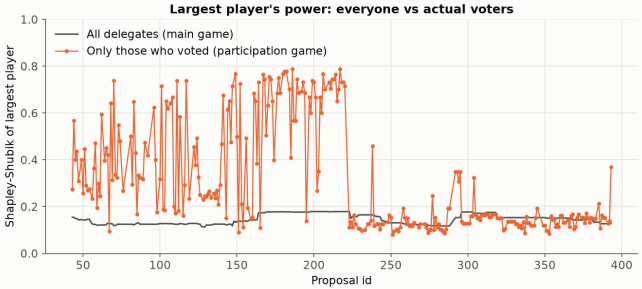
Median turnout: 25.5% of delegated weight. Low turnout is what makes a few large delegates decisive.



Beyond the base game: power among those who actually vote

Players = the **for/against voters** only; same quota rule on votes cast.

- Quorum (400k COMP) now binds in 259 of 314 decided proposals.
- In 23, the voters could not reach quorum at all.
- Median: 21 voters; the largest holds 17.6% of cast votes but 18.3% Shapley–Shubik power.
- Nakamoto (Shapley–Shubik) drops from 9 to 4.



Agreement with prior work

We recomputed the published Compound metrics on our own data.

Source	Metric	Published	Ours
Fritsch, Müller & Wattenhofer (2022)	Nakamoto coefficient of delegates, block 14.3M	8	**8**
Fritsch, Müller & Wattenhofer (2022)	Gini coefficient of delegates, block 14.3M	0.987	**0.986**
Messias et al. (2023)	mean voters needed for $\geq$50% of votes cast	3.18	**3.55**

- Ours: snapshot block 14,252,261; 314 decided Bravo proposals. Messias et al. also include Governor Alpha and data to Aug 2024.
- **What we add:** neither paper computes Banzhaf or Shapley–Shubik indices. Fritsch et al. use "could one delegate flip this vote"; we compute both indices exactly over the delegation-resolved game, for every proposal.

Their power measure vs our indices

Potential power (Fritsch et al., Def. 4.1): could this voter, by switching its vote, have flipped the result? **Exercised power** (Def. 4.2): did removing its vote flip it?

On our data (Compound rule, with quorum)

- Only 34 of 2,338 voters *ever* had potential power, in 96 of 314 proposals.
- One delegate could have flipped 57 of the 84 proposals it voted on.
- Sanity check: every such proposal also has a one-address flip set in our analysis (asserted in code).

Why we need full indices

- Potential power checks a swing at *one* coalition, the one that happened. Banzhaf averages swings over *all* 2^{n-1} coalitions; Shapley–Shubik over all orderings.
- It is 0 for most voters in most votes, so it cannot rank them; our indices can.
- Yet they agree: potential-power rate vs our Shapley–Shubik, Spearman 0.70 over 552 voters with $\geq$5 votes.

Takeaways

1. **Concentration is real, and slightly worse than token share suggests.** The top delegate's power exceeds its share under both indices in 100.0% of proposals; the median Nakamoto coefficient falls from 10 (weight) to 9 (Shapley–Shubik).
2. **The two indices agree on who matters** (by theorem) and differ by only $\sim$0.015 in total variation, but Banzhaf amplifies the top delegate more.
3. **Among actual voters it is starker:** quorum binds in 259 of 314 proposals and 4 voters hold most of the power.
4. **Realised outcomes are fragile.** The top 5 delegates could have reversed 97.5% of all decided proposals; a single address, 118 of 314.

Limitations

- **Snapshot choice:** weights at `startBlock`, exactly as the protocol counts; delegation can change during voting but cannot affect votes.
- **Delegation rule:** one-hop, as in COMP. A transitive reading moves at most 5.3% of weight in any proposal.
- **Addresses $\neq$ people:** one entity may control many addresses, which would make concentration *higher*. Fritsch et al. cluster addresses by transfers but call it "not bullet-proof"; we do not cluster.
- **Model assumptions:** the index game assumes all players vote and are not strategic (no coordination). Real turnout is 25.5%; the participation game (slide on actual voters) relaxes the all-vote assumption.
- **Rounding:** integer weights at $W/20000$; effect ≤ 0.00016 on any index.
- **Scope:** one DAO and one governor contract (GovernorBravo, 2021–2025).

Future directions

- **Turnout models:** between our two games, weight players by their historical participation.
- **Entity clustering:** merge addresses controlled by the same organisation.
- **Second DAO:** Uniswap uses the same token/governor design, so the pipeline transfers with a config change.
- **Off-chain voting** (Snapshot) and liquid-delegation designs where chains really are transitive.
- **Mechanism design:** how quorum or vote caps would change the indices.

References: Fritsch, Müller & Wattenhofer, *Analyzing Voting Power in Decentralized Governance: Who controls DAOs?*, arXiv:2204.01176.

Messias et al., *Understanding Blockchain Governance: Analyzing Decentralized Voting to Amend DeFi Smart Contracts*, arXiv:2305.17655.

Thank you

Questions?

Code, data pipeline and all results: `README.md` section 10 reproduces every number.